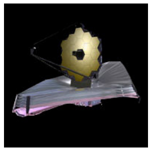

11286 - The Euclid–JWST Synergy: Revealing the Nature of Extremely Luminous z~9 Galaxies

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Ms. Natalie Pearl Allen (PI) (ESA Member)	**University of Copenhagen, Niels Bohr Institute**
Prof. Pascal Oesch (CoI) (ESA Member) (CoPI) (Contact)	University of Geneva, Department of Astronomy
Dr. John R. Weaver (CoI) (CoPI) (Contact)	Massachusetts Institute of Technology
Prof. Sune Toft (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Rebecca A A Bowler (CoI) (ESA Member)	University of Manchester
Mr. Rohan Gadiyar Varadaraj (CoI) (ESA Member)	University of Oxford
Dr. Hakim Atek (CoI) (ESA Member)	CNRS, Institut d'Astrophysique de Paris
Prof. Christopher Conselice (CoI) (ESA Member)	University of Manchester
Mr. Thomas Harvey (CoI) (ESA Member)	University of Manchester
Dr. Bo Milvang-Jensen (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Ms. Aryana Haghjoo (CoI)	University of California - Riverside
Prof. Stephen Serjeant (CoI) (ESA Member)	Open University
Dr. Conor McPartland (CoI)	University of California - Riverside
Dr. Adriano Fontana (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Prof. David B. Sanders (CoI) (US Admin CoI)	University of Hawaii
Dr. Yuichi Harikane (CoI)	University of Tokyo, Institute of Cosmic Ray Research
Grey Marie Murphree (CoI)	University of Hawaii at Manoa

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
	1		NIRSpec IFU Spectroscopy	(1) EDFN_1
	2		NIRSpec IFU Spectroscopy	(2) EDFN_2
	3		NIRSpec IFU Spectroscopy	(3) EDFN_6
	4		NIRSpec IFU Spectroscopy	(4) EDFN_7
	5		NIRSpec IFU Spectroscopy	(5) EDFN_5
	6		NIRSpec IFU Spectroscopy	(6) EDFN_3
	7		NIRSpec IFU Spectroscopy	(7) EDFN_4

ABSTRACT

The first billion years of cosmic history represent a critical epoch when the very first sources of light emerged. While JWST has made great progress to probe cosmic dawn, our understanding of early galaxy assembly requires us to probe both the faint, numerous and the rare, luminous sources. With JWST, we remain limited by small survey volumes. Luckily, the new ESA/Euclid mission is about to revolutionize this field. It allows us to probe a cosmic volume >100x larger than current JWST surveys, providing the first statistically robust constraints on rare, extremely luminous galaxies at cosmic dawn. Here we exploit the first data over 10deg2 of the Euclid Deep Field North to identify an unparalleled sample of seven z~9 galaxies with Muv~-23, i.e. >2mag brighter than any JWST-selected galaxies at similar redshifts. We propose to reveal the nature of these unique sources with short NIRSpec/IFU spectroscopy. This will provide their spectroscopic redshifts, accurate stellar masses, and ISM conditions, while simultaneously measuring their morphologies and multi-clump structures, as well as a possible AGN contribution that could explain their extreme luminosities. The strong synergy between Euclid's unparalleled wide-area view and JWST's unmatched spectroscopic capability in the near-infrared open up a completely new parameter space to stress-test our models of early galaxy formation at the extreme end of the galaxy population.

OBSERVING DESCRIPTION

This proposal focuses on a sample of 7 ultra-luminous z~9 galaxies selected from the first 10 deg2 of Euclid/Deep imaging. We will make use of NIRSpec IFU spectroscopy to confirm the redshifts of these galaxies, and characterize their physical properties.

The high luminosity of these sources (>2mag brighter than GN-z11) allows us to reach high a continuum S/N of >5 per spectral element in 20min on-source time. This SN further enables us to spatially dissect these galaxies and derive resolved physical parameters from strong line metallicities to UV slopes and Balmer breaks.

We will use the NRSIRS2RAPID readout pattern with a LARGE SPARSE-CYCLE DITHER with positions 1-4 to allow for optimal background subtraction.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	EDFN_1	RA: 17 44 9.6686 (266.0402858d) Dec: +66 18 56.92 (66.31581d) Equinox: J2000		
Comments: Category=Galaxy Description=[Lyman-break galaxies]				
(2)	EDFN_2	RA: 17 51 49.7222 (267.9571758d) Dec: +65 13 25.88 (65.22386d) Equinox: J2000		
Comments: Category=Galaxy Description=[Lyman-break galaxies]				
(3)	EDFN_6	RA: 18 01 2.2521 (270.2593837d) Dec: +64 45 16.84 (64.75468d) Equinox: J2000		
Comments: Category=Galaxy Description=[Lyman-break galaxies]				
(4)	EDFN_7	RA: 18 11 31.2440 (272.8801833d) Dec: +66 48 57.70 (66.81603d) Equinox: J2000		
Comments: Category=Galaxy Description=[Lyman-break galaxies]				
(5)	EDFN_5	RA: 18 00 47.6237 (270.1984321d) Dec: +66 02 49.82 (66.04717d) Equinox: J2000		
Comments: Category=Galaxy Description=[Lyman-break galaxies]				
(6)	EDFN_3	RA: 17 52 4.9834 (268.0207642d) Dec: +64 50 39.13 (64.84420d) Equinox: J2000		
Comments: Category=Galaxy Description=[Lyman-break galaxies]				
(7)	EDFN_4	RA: 17 59 3.9486 (269.7664525d) Dec: +64 27 33.82 (64.45939d) Equinox: J2000		
Comments: Category=Galaxy Description=[Lyman-break galaxies]				

Observation

Proposal 11286, Observation 1

Fri Mar 13 21:03:39 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	EDFN_1	RA: 17 44 9.6686 (266.0402858d) Dec: +66 18 56.92 (66.31581d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Lyman-break galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	SPARSE-CYCLING	LARGE			1, 2, 3, 4

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	20	1	false	true	NONE	4	4	1225.467	

Observation

Proposal 11286, Observation 2

Fri Mar 13 21:03:39 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	EDFN_2	RA: 17 51 49.7222 (267.9571758d) Dec: +65 13 25.88 (65.22386d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Lyman-break galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	SPARSE-CYCLING	LARGE			1, 2, 3, 4

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	20	1	false	true	NONE	4	4	1225.467	

Observation

Proposal 11286, Observation 3

Fri Mar 13 21:03:39 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	EDFN_6	RA: 18 01 2.2521 (270.2593837d) Dec: +64 45 16.84 (64.75468d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Lyman-break galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	SPARSE-CYCLING	LARGE			1, 2, 3, 4

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	20	1	false	true	NONE	4	4	1225.467	

Observation

Proposal 11286, Observation 4

Fri Mar 13 21:03:39 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	EDFN_7	RA: 18 11 31.2440 (272.8801833d) Dec: +66 48 57.70 (66.81603d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Lyman-break galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	SPARSE-CYCLING	LARGE			1, 2, 3, 4

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	20	1	false	true	NONE	4	4	1225.467	

Observation

Proposal 11286, Observation 5

Fri Mar 13 21:03:39 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	EDFN_5	RA: 18 00 47.6237 (270.1984321d) Dec: +66 02 49.82 (66.04717d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Lyman-break galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	SPARSE-CYCLING	LARGE			1, 2, 3, 4

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	20	1	false	true	NONE	4	4	1225.467	

Observation

Proposal 11286, Observation 6

Fri Mar 13 21:03:39 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	EDFN_3	RA: 17 52 4.9834 (268.0207642d) Dec: +64 50 39.13 (64.84420d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Lyman-break galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	SPARSE-CYCLING	LARGE			1, 2, 3, 4

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	20	1	false	true	NONE	4	4	1225.467	

Observation

Proposal 11286, Observation 7

Fri Mar 13 21:03:39 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(7)	EDFN_4	RA: 17 59 3.9486 (269.7664525d) Dec: +64 27 33.82 (64.45939d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Lyman-break galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	SPARSE-CYCLING	LARGE			1, 2, 3, 4

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	20	1	false	true	NONE	4	4	1225.467	